

YOLO plant diseases detection

Objective

Its to create a YOLO based object detection model for detecting plant diseases, Which will help in identifying the diseases early on so that the plants could be saved and promote them to grow more healthier which will result in good plant yields. The model was trained with a pre made dataset which is available on kaggle called [PlantDoc](#). Its a relatively small dataset with only 2569 images of plant leaves with diseases. The model was then tested on images outside the dataset to test it's accuracy.

Methodology

The dataset was acquired from Kaggle which already had the images pre annotated which helped me in training the model easier. The pre annotated images were split into Train, Test, and Validation sets using a `data.yaml` file the dataset paths were mentioned along with the prediction classes for the images. Then the **YOLO8n** model was loaded from ultralytics library to which the data.yaml file was passed as parameters. The training was done for 80 epochs.

Results

The trained YOLOv8 model achieved a Precision of approximately **54%**, Recall of **54%**, mAP@50 of **54%**, and mAP@50-95 of **37%**. The training and validation loss curves indicate successful convergence with only minor signs of overfitting during the later epochs

- **Precision:** ~0.54
- **Recall:** ~0.54
- **mAP@50:** ~0.54
- **mAP@50-95:** ~0.37

Conclusion

The trained model achieved satisfactory performance, with a Precision of approximately **54%**, Recall of **54%**, **mAP@50** of **54%**, and **mAP@50-95** of **37%**, indicating its ability to detect and classify plant diseases with reasonable accuracy. The project provided valuable hands-on experience in dataset preparation, model training, evaluation, and deep learning based object detection. Future work can focus on improving accuracy by using a larger and more diverse dataset, applying data augmentation techniques, and optimizing the model for real-time deployment in agricultural applications.

